

Position Specific Scoring Matrix and Synergistic Multiclass SVM for Identification of Genes

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Abstract—In genome annotation field several methods have been developed to locate the patterns of genes in genome sequence. In this paper we propose a novel method for identifying genes from protein sequences. The first step of the proposed method involves computing a position specific scoring matrix (PSSM) of protein sequences. The normalized PSSM is used to convert protein sequences into training data set. The data set is simplified by averaging the normalized values corresponding to the same amino acid that occur at more than one location of the protein sequence. The resulting training data set is used to train multiclass Support Vector Machine (SVM) classifier that relates simplified normalized values of amino acid to various genes. The results of several multiclass SVMs are synergistically combined for improving the results. The proposed approach is tested on genome DNAs dataset. Empirical evaluation shows that the proposed new approach produces good results of identifying genes using protein sequences.

Keywords—Gene Identification, PSSM, SVM, Synergistic Multiclass Classifier.

I. INTRODUCTION

Given a genome, the most important task is to determine the structure of protein-coding genes, representing the majority of the transcribed and the translated genes. They show incredible diversity in size and organization but have some conserved features.

The central part of the gene is the coding sequence (CDS) that consists of the nucleotides translated in the protein amino acids. The initial codon, represented by ATG, is the fragment with coding genes and the ending fragment is represented with one of three termination codons: TAA, TAG or TGA. The part before the coding region and after the coding region consists of DNA sequences which are called as exons. Exons may be interrupted by introns, DNA sequences that are cut in the mature transcripts by a splicing process to form the messenger RNA (mRNA).

DNA binding proteins plays an essential role in all cell functions such as DNA replication, DNA repair, DNA modification and all the other activities allied with DNA. Most of the genes include statistics for generating proteins at definite level and these proteins are then used to carry out a broad diversity of procedures in the unit. Other type of genes, known as non-coding genes, determines efficient genetic

material (RNA) which is occupied in the guidelines of appearance of genes and production of proteins. This sequence of DNA is not allowed to transform into amino acids and hence be deficient in the distinctive sequence restriction of coding sequences.

The gene identification from large genome sequence is found to be one of the important research problems in the field of bioinformatics. The primary issue in the process of gene identification is to locate the protein coding genes in genomic DNA sequence. In spite of large amount of amino acid sequences of proteins produced, only a small part of protein function has been interpreted.

The big data associated with the problem of identification of genes can be projected into sub-spaces or clusters so that the given problem can be divided into sub-problems. Each sub-problem can then be optimized with an appropriate model. One of the approaches of dividing a given problem into sub-problems involves projecting the big data to various sub-space grids where each sub-space grid represents a sub-problem [1-3]. Each sub-problem can then be represented independently with various models which can be combined by using a rule-based system [4]. In this work we explore using new approach for identification of genes using protein sequences by employing position specific scoring matrix and synergistic multiclass support vector machine [6, 21].

The remaining portion of the paper is organized as follows: In section II, we first give the brief review of gene finding approaches. Section III explains the proposed approach for gene identification. Section IV describes the dataset used and provides the evaluation of this approach. Finally, conclusion is given in section V.

II. LITERATURE REVIEW

The specific recognition of genes is one of the elementary steps in all meta-genomic sequencing projects [5]. The functional proteins included in organisms at the upper level are not adjacent. These are frequently divided into coding and non-coding regions. Such types of coding fragments are recognized as exons. The exonic sequences are then mixed together by non-coding section of exceedingly changeable length known as introns. The extensively used move toward the genome annotation

consists of two methods namely extrinsic and intrinsic methods. The extrinsic methods are used for homology detection [7] and intrinsic methods are used for gene identification [8]. The homology methods when executed can forecast only half portion of the genes and the rest of the genes remain unknown. Therefore, more extrapolative, fast and reliable methods are needed which can detect all the protein coding genes accurately. The method of assimilating nucleic acid similarity search has been shown practically in a long line of accomplishment, including GRAIL [9], HMMgene [10], and Genscan [11] and GenomeScan[12].

The most important challenge that follows the sequencing of either a small segment of DNA sequence or a long genome sequence is to establish the location of functional units like protein coding genes (exons), splice sites, terminators etc. This provides a procedure of identifying the regions that encode proteins. The protein homology detection also takes an account of recognizing the patterns in multidimensional data. This type of segment is known as an Open Reading Frame (ORF) [13] which assembles like a gene but it has not been proved to be a gene yet.

The gene detection techniques are applied to identify the gene catalog in a genome sequence. It includes several phases: 1) identification of gene positions and localizations on the genome, 2) determination of exon-intron boundaries and 3) discovery of molecular signals (e.g. start/stop codons, splicing sites, etc.). In genome projects, these tasks are accomplished through several computational methods that show great reliability in discovering structure signals and reconstructing gene boundaries. The three major computational approaches are ab-initio gene finders, ESTs or protein alignments and comparative genomics.

A number of gene finding approaches have been proposed by researchers and can be broadly classified into three categories: a) finding open reading frames (ORF), b) homology based approach which involves pair-wise alignment and c) ab-initio approach.

A. Gene Finding Using ORF Based Approach

In case of prokaryotes, gene recognition is relatively dissimilar problem as compared to gene recognition in eukaryotic genomes. The prokaryotic genome involves the two properties of having higher gene density and the absence of introns in their protein coding genes. That portion of coding genes are called as open reading frames (ORFs) which may be longer is likely correspond to genes. The main problem which arises in this approach are frame-shift errors, that is a single insertion or deletion might lead to completely dissimilar amino-acid; very small genes may get miss; overlapping of long ORFs on opposite strands of DNA may leads to ambiguity.

B. Homology Based Approach

Homology based approaches are categorized into two types for gene identification: the comparison of the DNA query sequence with a protein or cDNA sequence, or a database of such sequences, and the comparison of two or more genomic sequences. In both approaches query and

target sequences may be from the same or different species. In these methods, genes can be predicted by aligning input sequences to the nearby homologous sequences in the database.

C. Ab-Initio Based Approach

The ab-initio system is one of the gene-finding tools that has been used by researchers [14]. An advantage of ab-initio gene finders is that the detection of gene evidences which cannot be discovered through other means such as expression data. These methods instruct various factors on recognized annotations so that unidentified annotations can be predicted.

GenScan [11] has been used for recognition of genes by employing a multifaceted probabilistic representation of the gene composition. The organization of the genomic data is modeled by an unambiguous state duration using Hidden Markov Model. GenomeScan [12] has been used to identify the coding and non-coding regions of genes in genomic DNA chain of characters. GeneID [15] was used to recognize full coding gene configuration of vertebrate genes in unstipulated DNA sequences. GeneMark has been used for finding prokaryotic genes [16]. This tool has been used to engage a non standardized Markov model to organize DNA sections into protein coding and non-coding. This process has been modified very recently to recognize gene composition in eukaryotic living beings. FGENESH has been used for fast and precise gene prediction [17]. Another gene finding approach known TwinScan is an extension of GenScan approach where a comparison is made between the query and target sequences so as to either increase or decrease the probability of finding a gene. The query sequence and target sequences are compared to produce one of the three results of match, mismatch or unaligned sequences.

Position-specific iterated BLAST (PSI-BLAST) is the most frequently utilized program, which draws a comparison of PSSM profiles for identifying distantly related homogeneous proteins or DNA [18, 19]. The rudimentary PSSM given in [20]. The work described in this paper explores how PSSM profiles can be used for identifying genes.

III. PROPOSED APPROACH FOR IDENTIFYING GENES

In this paper, we propose a new approach of identifying genes using protein sequences. The proposed approach has the following steps:

- i) A normalized Position Specific Scoring Matrix (PSSM) of protein sequences is computed.
- ii) Each protein sequence is labelled with an appropriate gene name that it represents.
- iii) The protein sequences are transformed into training dataset using the normalized PSSM.
- iv) Synergistic multiclass SVM classifier is trained with the training dataset obtained above.

The above steps are explained in detail below.

A. PSSM of Protein Sequences

A position specific scoring matrix (PSSM) of protein sequences is a matrix that contains probability of occurrence of each type of amino acid at each position of protein sequence. Computation of PSSM results in a matrix of dimension $20 \times L$ (20 rows and L columns), where rows represent 20 amino acids and L represents the maximum length of protein sequences.

The normalized PSSM profile of protein sequences is created using the following steps:

Step 1: The number of times an amino-acid ' i ' appears at column ' j ' is calculated and denoted by n_{ij} .

where ' i ' represents amino-acids ranging from 1 to 20 and ' j ' represents the sequence index ranging from 1 to L (maximum length of a protein sequence).

Step 2: The frequency of amino acid ' i ' at position ' j ' as a ratio of total number of sequences is calculated as

$$F[i, j] = \frac{n_{ij}}{k}$$

where ' k ' is the total number of sequences.

Step 3: Assuming that every amino acid appears equally in protein sequences, i.e. $f(i) = 1/20 = 0.05$ for every amino acid ' i ', the PSSM scores are calculated by taking the logarithm of the values obtained above divided by the probability of an amino acid occurring in the sequence as shown below:

$$Score(i, j) = \log[F(i, j)/f(i)]$$

where $F(i, j)$ is the frequency of amino acid ' i ' observed at position ' j ' and $f(i)$ is the overall frequency of amino acid ' i '.

Step 4: Normalize the values of the PSSM score in range of 0–1 by using the formula:

$$PSSM = (Score(i, j) - MinVal) / (MaxVal - MinVal)$$

where $MinVal$ and $MaxVal$ represent the minimum and maximum values of $Score(i, j)$ respectively.

B. Labelling of Protein Sequences with Gene Names

Each protein sequence is labelled with appropriate gene name that it represents using BLAST tool. The protein sequence is given as an input to BLAST tool (BlastP, in case of proteins) [21]. The output generated consists of three fields: i) the header which contains information about the input protein sequence and a summary of database searched, ii) the one line description of each database sequence when matched to input protein sequence which includes Accession number of matching sequence, alignment score in bits and a statistical significance, iii) the matched alignments for each sequence and its gene name.

C. Transforming Protein Sequences into Training Data Set

The protein sequences are transformed into training data set using the steps given below.

Step 1: Each amino acid in a protein sequence is replaced by the corresponding normalized PSSM value. This normalized PSSM value of each amino acid is retrieved by mapping the index of an amino acid in a protein sequence with the corresponding index in the PSSM.

Step 2: Replace the normalized values of same amino acids in the transformed protein sequence by its average value (by adding normalized values of an amino acid at various locations and dividing it by the number of times the amino acid repeats in the sequence). This reduces the size of the transformed training data vector of each protein sequence to 20 (i.e. number of amino acids).

Step 3: Arrange the averaged normalized values of the transformed data vector according to the following order of amino acids: (ARNDCQEGHILKMF-PSTWYVBZ). This is repeated for all protein sequences to generate normalized training data set.

To improve the performance of classifiers, Protein sequences are divided into a number of groups, where each group related to a few genes. The training data set is obtained for all groups and trained on classifiers independently.

D. Synergistic Multiclass SVM Classifier

We propose using synergistic multiclass SVM classifier for training purpose. Combining multiclass SVM classifiers synergistically improves the performance of identifying genes correctly.

The synergistic multiclass SVM classifier is shown in Figure 1 below. It uses three SVM modules: DAGSVM, BTSVM, and BDT SVM. The output from the three modules is synergistically combined to produce the final output.

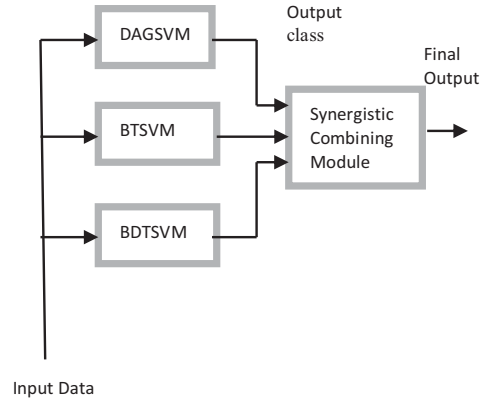


Figure 1. Synergistic Multiclass SVM Classifier

IV. RESULTS AND DISCUSSION

The dataset used here is gene DNAs et that is used in [22]. The gene DNAs et dataset is protein sequence of 20 attributes (20 amino acids): A(Alanine), R(Arginine), N(Asparagine), D (Aspartic Acid), C(Cysteine), E(Glutamic Acid), Q (Glutamine), G(Glycine), H(Histidine), I (Isoleucine), L(Leucine), K(Lysine), M (Methionine), F(Phenylalanine), P(Proline), S(Serine), T (Threonine), W(Tryptophan), V(Valine) and Y (Tyrosine). An amino acid may occur in a protein sequence at more than one location.

A total of 2435 DNA-Binding proteins from Protein Data Bank (PDB) were extracted and proteins having peak similarity with other proteins were filtered. This resulted in 146 non-redundant DNA-binding proteins and 250 non-redundant non-binding proteins.

TABLE I. Synergistic Multiclass SVM Approach for Identifying genes in 30 groups of protein Sequences

Protein sequences group number	Number of protein sequences per group	Accuracy rate of identifying genes
1	37	100
2	16	100
3	12	100
4	23	95.6
5	5	80.0
6	8	87.5
7	12	100
8	5	100
9	7	85.7
10	23	95.6
11	6	83.3
12	6	83.3
13	9	100
14	9	100
15	10	90
16	4	100
17	14	100
18	8	100
19	3	100
20	27	96.3
21	19	94.7
22	6	83.3
23	36	100
24	32	81.3
25	6	83.3
26	12	91.6
27	18	83.3
28	11	100
29	6	100
30	6	100

The protein sequences are divided into 30 groups by assembling the protein sequences with same gene id's into a single group where each group has sequences that correspond to a few genes. The normalized PSSM is used to transform protein sequences into training data. Synergistic multiclass SVM is trained to identify genes using training data set. Table I shows the accuracy of identifying the genes using the proposed approach.

The Table I shows that the proposed approach of identifying genes from protein sequences by employing normalized PSSM and synergistic multiclass classifier produces good results.

V. CONCLUSION

In this paper a new PSSM based approach was proposed for identifying genes present in a given protein sequence. The new approach generated PSSM profile for given protein sequence. The PSSM profile was used for transforming protein sequences into training data set. Synergistic multiclass SVM classifier was proposed for identifying genes. **This new approach was tested on DNAs et dataset for identifying the genes.** The experimental results indicate that the proposed method produced good result.

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